Session 1: Foundations of Marine Macroecology and Niche Theory

Modeling the distribution of biodiversity under global climate change

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Unpacking the Ecological Niche

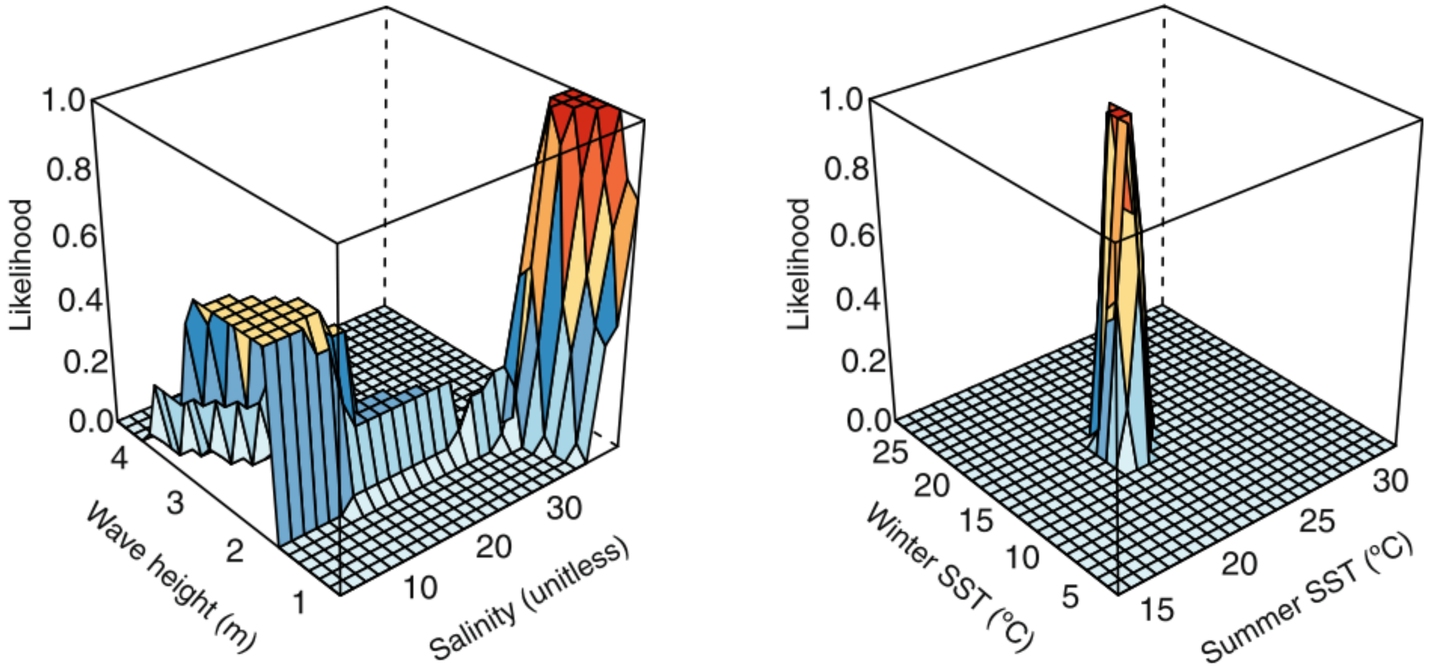
The ecological niche: an evolving core concept

A foundational concept in Macroecology. Evolved through different perspectives:

Grinnellian niche (Joseph Grinnell, 1924): the abiotic conditions (e.g., temperature tolerance) and habitat requirements (e.g., substratum or type) that determine where a species can survive and reproduce.

Eltonian niche (Charles Elton, 1927): the species' functional role within its community, in particularly the interactions it has with other organisms (e.g., competition, predation).

Hutchinsonian niche (G. Evelyn Hutchinson, 1957): a **"n-dimensional hypervolume”** that includes the **abiotic** conditions (i.e., environmental), the **biotic** interactions (e.g., availability of prey) and **resource** requirements (e.g., availability of nursing sites) under which **a species can maintain viable populations** (i.e., survival and reproduction).



Hypervolume of abiotic conditions (environmental dimensions) where a Mediterranean coral can survive and reproduce (fundamental niche).

Boavida et al, 2016

Fundamental vs. realized niche

Hutchinson also distinguished between:

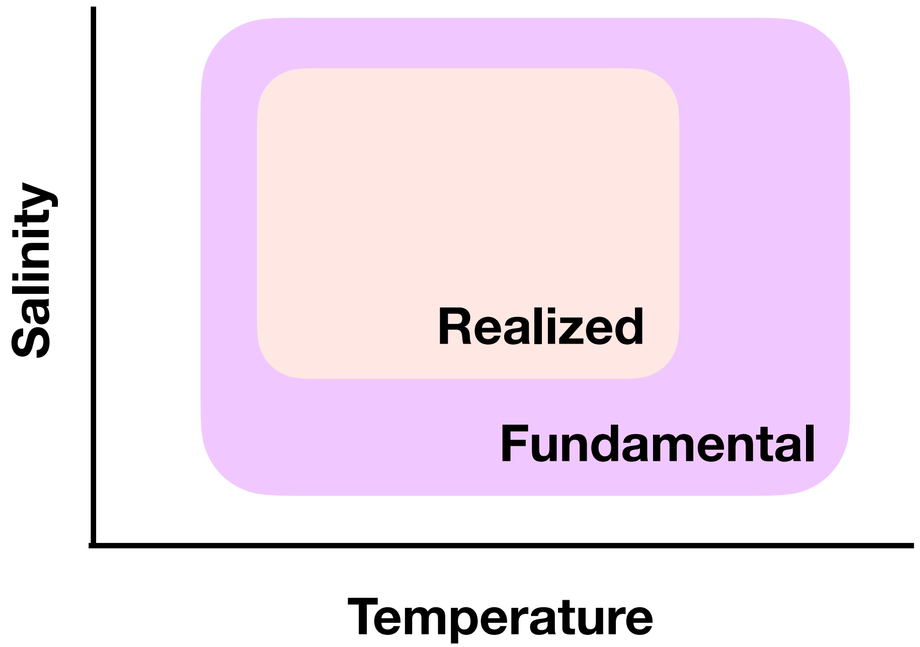
Fundamental niche: the **full range of environmental conditions and resources** (dimensions in the hypervolume) allowing a species to potentially survive and reproduce indefinitely, **in the absence of biotic interactions**.

Realized niche: the **portion of the fundamental niche** (a subset of the hypervolume dimensions) **that a species actually occupies in the presence of limiting biotic interactions** - competition with other species for resources, or predation pressure restricts a species from utilizing its full fundamental niche.

Fundamental vs. realized niche

The fundamental niche is the theoretical maximum niche space the species can occupy.

Fundamental Niche > Realized Niche



Are niches fixed? Conservatism vs. evolution

A central question in Macroecology: How stable are species' ecological niches?

Two opposing perspectives:

Niche conservatism: niches remain stable over time.

Niche evolution: niches can evolve over time.

The balance between these perspectives influences species’ distribution, their local adaptation, and speciation.

Niche conservatism

The tendency for **species to retain ancestral niche characteristics** over long periods (e.g., thermal tolerance).

Implications:

Implies **no adaptive or evolution potential** to novel conditions (fixed or conserved).

Species directly **respond to climate change by going extinct or by shifting their geographic distribution** (tracking suitable conditions), rather than adapting in place.

Evidence: observed distribution shifts match climate shifts. Without niche conservatism, species would persist locally while climate conditions shift.

Niche evolution

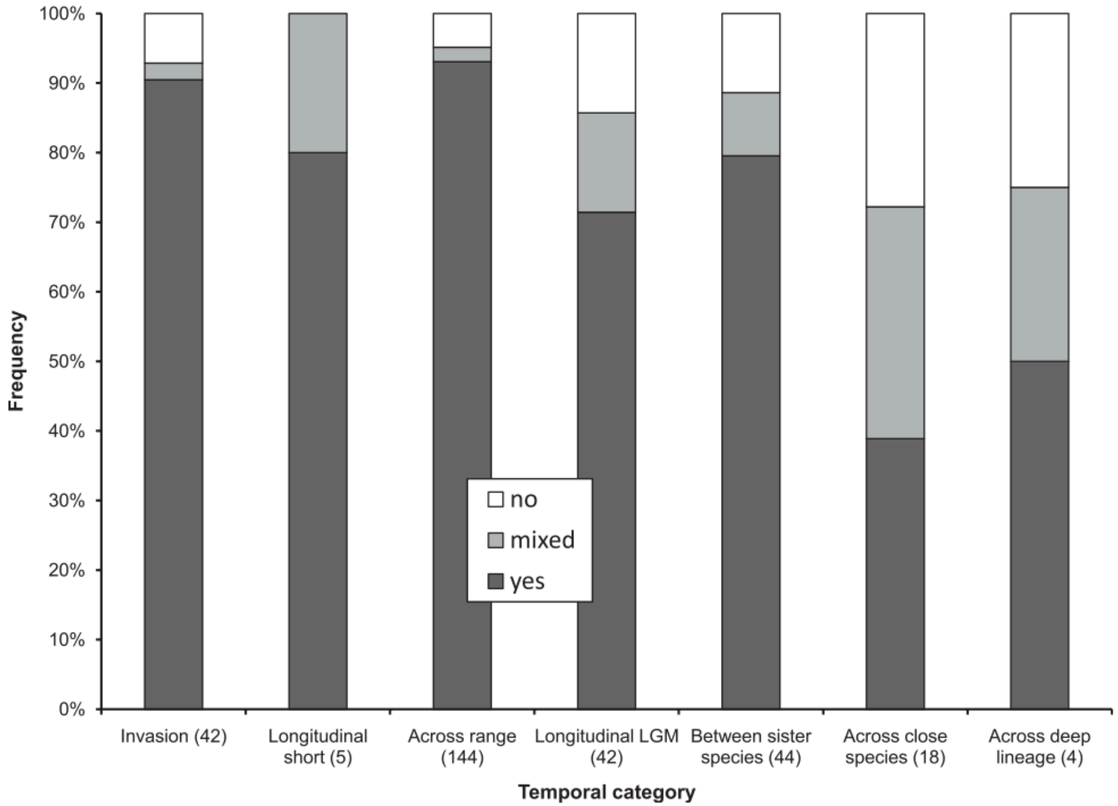
Niches can evolve, allowing species to colonize new environments.

Happens through natural selection acting on physiological tolerances, resource use, etc.

The key question: is niche evolution (i.e., adaptation potential) fast enough to keep the pace of the rapid, human-induced environmental change?

If evolution is slow relative to climate change, niche conservatism dominates responses (e.g., local extinctions and range shifts).

If fast enough, adaptation is possible under ongoing climate change (e.g., persistence).



An extensive review **shows evidence for ecological niches being highly conserved from short to deep time spans.**

(i.e., from individual life spans - invasions - up to hundreds of thousands of years).

Even at deep geological time scales of thousands of years NC of ~50% of species.

Peterson et al., 2011